08-09 第1篇

阅读理解（说明文） How Does a Thermos Keep Drinks Hot (and Cold)?

① On a winter morning, your mother fills a small metal bottle with hot soy milk. You carry it in your bag for three or four hours, and at lunch time you open it — the soy milk is still warm, and small clouds of steam rise from the cup. In summer, the same bottle is used for ice water, and six hours later the water is still cold enough to make your hand feel cool. A simple bottle, nothing electric, no switch, no battery. **How does a thermos keep hot drinks hot and cold drinks cold at the same time?**

② To answer this, we need to remember a basic idea from science class: **heat always moves from hotter places to cooler places**. A hot drink in a cool room loses heat to the air; cold water in a warm room gains heat from the air. To keep a drink's temperature steady, we must **slow down this movement of heat as much as possible**. There are three main ways that heat can travel: **conduction**, **convection**, and **radiation**. A good thermos is clever because it blocks all three.

③ A modern thermos is, in fact, **a bottle inside a bottle**. Both inner and outer bottles are made of a thin layer of stainless steel or glass. Between them is a **narrow space**. From the outside, a thermos looks like one simple container, but inside its walls, something special has been done with this middle space: almost all of the air has been pumped out, creating a **vacuum** — that is, a space with very few air molecules.

④ Why does the vacuum matter? Because air is one of the main carriers of heat. In normal life, hot air near a cup rises and is replaced by cooler air below, and so on; this moving flow of air is called **convection**. Molecules of air also bump into each other and pass heat from one to the next — this is **conduction**. If there is almost no air in the middle space of the thermos, there is almost nothing there to carry heat between the two walls. Conduction and convection across that space are therefore very small.

⑤ But there is a third way heat can travel: **radiation** — heat moving as invisible waves, the way sunlight reaches us through the empty space of the solar system. Even a perfect vacuum cannot block radiation. To fight this, the inner wall of a thermos is often coated with a **shiny silver layer**. Shiny surfaces are poor at sending heat out and poor at letting heat in; they tend to **reflect** heat waves back to where they came from. So the hot soy milk's heat waves mostly bounce back into the drink, and the warm air of a summer day mostly bounces off the outside of the bottle.

⑥ The last small enemy of a thermos is the **lid**. Air under the lid can carry heat out, and if the lid does not seal well, steam can escape. That is why a good thermos has a **thick**

rubber or plastic seal inside the lid — when it is screwed tight, almost no air gets in or out through the top.

⑦ So the trick of the thermos is not a "magic secret"; it is actually **several simple ideas working together**: a vacuum to block conduction and convection; a shiny surface to reduce radiation; a well-sealed lid to stop warm air or steam from escaping. None of these tricks is very advanced by itself. Their power is in the combination.

⑧ A bit of history: the first vacuum flask was invented in 1892 by a Scottish scientist named Sir James Dewar. He was studying very cold liquid gases in his laboratory, and he needed a container that would keep them cold long enough to study. The flask he invented was originally a laboratory tool. Years later, a German company began to sell it as an everyday drink bottle and called it "Thermos". Today, a design meant for scientific glass tubes sits quietly in millions of schoolbags.

⑨ Knowing how a thermos works also teaches us good habits. To keep water hot longer, **pre-warm** the thermos: fill it with hot water, wait one minute, pour it out, then add the real hot drink. This way, the inner wall will not steal heat from the drink at first. To keep cold drinks cold, do the same with cold water. Never leave the lid loose, and never store coffee in a thermos for more than twelve hours — hot coffee can slowly change its taste when kept in a metal bottle.

⑩ A thermos is a small, quiet example of how science helps us in daily life. A hundred years ago, it was a laboratory tool; today, it keeps your mother's kindness warm in your school bag until lunch. Behind that everyday warmth is a vacuum, a shiny coat, and a tight lid — three simple inventions, working together, without any battery at all.

根据文章内容，选择最佳答案。

1. According to the passage, which is the basic scientific idea behind a thermos?

A. Cold objects always get colder by themselves.
B. Heat always moves from hotter places to cooler places, and a thermos slows this movement down.
C. Heat moves only through electricity.
D. Hot and cold can never be kept in the same bottle.

2. What is the middle space between the two walls of a modern thermos like?

A. It is filled with heavy oil.
B. It is full of warm air.
C. Almost all the air has been pumped out, creating a vacuum.
D. It is filled with soft cotton to hold the bottle in place.

3. Why is the inner wall of a thermos often coated with a shiny silver layer?

A. To make the bottle look beautiful.
B. To block heat radiation by reflecting heat waves back.
C. To hold the lid in place.
D. To kill germs in the drink.

4. What happens if the lid of a thermos does not seal well?

A. The drink becomes more colourful.
B. Warm air or steam can escape, so the drink loses heat faster.
C. The outside of the bottle becomes icy.
D. The rubber seal will explode.

5. What is the main purpose of the last paragraph?

A. To advertise a new kind of thermos.
B. To connect the science of the thermos with a small but warm human idea — everyday kindness delivered without electricity.
C. To teach readers to make their own thermos at home.
D. To criticize old scientific inventions.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段
【解析】热量从高温→低温；保温瓶只是减缓。A/C/D 均错。故选 B。

2.
【答案】C
【题型】细节理解题
【定位】第③段
【解析】两层瓶壁之间抽成真空。A/B/D 均错。故选 C。

3.
【答案】B
【题型】细节理解题
【定位】第⑤段
【解析】反射热辐射。A/C/D 均错。故选 B。

4.
【答案】B
【题型】细节理解题
【定位】第⑥段
【解析】密封不好蒸汽/空气外泄 → 热量流失。A/C/D 均错。故选 B。

5.
【答案】B
【题型】主旨大意题/推理判断题
【定位】第⑩段
【解析】科学与日常温情的结合。A/C/D 均偏。故选 B。

词汇注释（超纲词）

- **conduction** [n.] 热传导
- **convection** [n.] 热对流
- **radiation** [n.] 热辐射
- **vacuum** [n.] 真空
- **flask** [n.] 烧瓶；保温瓶